BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Timed Paper 1 Practice

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What is the purpose of the Data Protection Act 2018?

2. Why indent code?
   A) To fix every logic error
   B) To replace all conditions
   C) To change false into true
   D) To show blocks clearly

3. "A comment always changes the value of the variable on the next line." Correct this. Explain why.

4. An indicator uses Q = NOT A. Give outputs for A=0 and A=1, then evaluate NOT(NOT A) for both values.

Today's objectives

Explain the core ideas of timed paper 1 practice using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Paper 1 timing
The OCR J277/01 paper has 80 marks in 90 minutes.

Show calculations
Write formula, substitute numbers and give final units.

Plan explanations
Match distinct points and justifications to mark tariff.

Track time
Leave opportunity to revisit unanswered questions without sacrificing required detail.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

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I Do - Worked model

Model a four-mark image-size calculation: 80 x 50 pixels at 8 bits/pixel.

- Count 80 x 50=4000 pixels.
- Multiply by eight: 32,000 bits.
- Divide by eight: 4000 bytes.
- Check that metadata/compression are excluded.

We Do - Guided practice

1. Plan a six-mark comparison of SSD and HDD for a traveller.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Convert a raw data size from bits to bytes.

```
sizeBits = 24000
sizeBytes = sizeBits DIV 8
print(sizeBytes)
```

You Do - Independent exam questions

Q2. State RAM volatility and its implication. [2 marks]

Q3. Calculate the largest eight-bit unsigned number. [2 marks]

Q4. Explain why a router is needed between separate networks. [3 marks]

Exit Ticket - True or False?

Units can be ignored as long as the final calculation looks plausible.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. It sets rules for using personal data lawfully and protects people's rights.

2. D - To show blocks clearly

3. It explains important intent or reasoning without misleading the reader.

4. (a) Q: 1, 0. Double NOT: 0, 1, restoring A.

I Do - Explain the model

Count 80 x 50=4000 pixels.

Multiply by eight: 32,000 bits.

Divide by eight: 4000 bytes.

Check that metadata/compression are excluded.

We Do - Model answers

1. Balance with higher cost per GB; link to travel needs.

Code First - Answer

24,000 bits / 8 = 3,000 bytes (3 kB using decimal units). Changing to 48,000 bits gives 6,000 bytes (6 kB). This is a unit calculation, not a complete file-size formula.

You Do - Self-mark

Q2: Volatile; active contents are lost without power.

Q3: 255.

Q4: Forwards packets using logical/IP routing, allowing communication between networks.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.